

SQL FUNDAMENTALS

Database & SQL Foundations

14 core concepts, explained and answered — the vocabulary you need before writing your first query.





CONCEPT 01

What Is Data?

DEFINITION

Data is raw facts and figures — numbers, text, dates, or measurements — recorded without context. On its own, a single value doesn't tell you anything useful.

RAW VALUES (NO CONTEXT)

- "92,000"
- "Dhaka"
- "2019-05-09"
- "Software Engineer"

DATA BECOMES USEFUL WHEN IT'S ORGANIZED

"Mason Jackson, Software Engineer in Dhaka, earns 92,000 and joined on 2019-05-09." — the same facts, structured into a record, become information you can act on. That structuring is exactly what a database does.



CONCEPT 02

What Is a Database?

DEFINITION

An organized collection of related data, stored electronically and structured so it can be efficiently accessed, managed, updated, and shared by many users at once.



Employees

Names, roles, salaries, hire dates



Departments

Team names, budgets, locations



Attendance

Daily check-in / check-out records

All stored together as one database — Northstar's hr_training database — so it can be queried as a whole.



CONCEPT 03

What Is a DBMS?

DEFINITION

A Database Management System is the software that creates, stores, secures, and gives controlled access to a database. It's the engine — the database is just the data it manages.

THE ANALOGY

Database = the library — the books themselves

DBMS = the librarian — organizes, retrieves, protects access

COMMON DBMS PLATFORMS

MySQL

PostgreSQL

SQL Server

Oracle

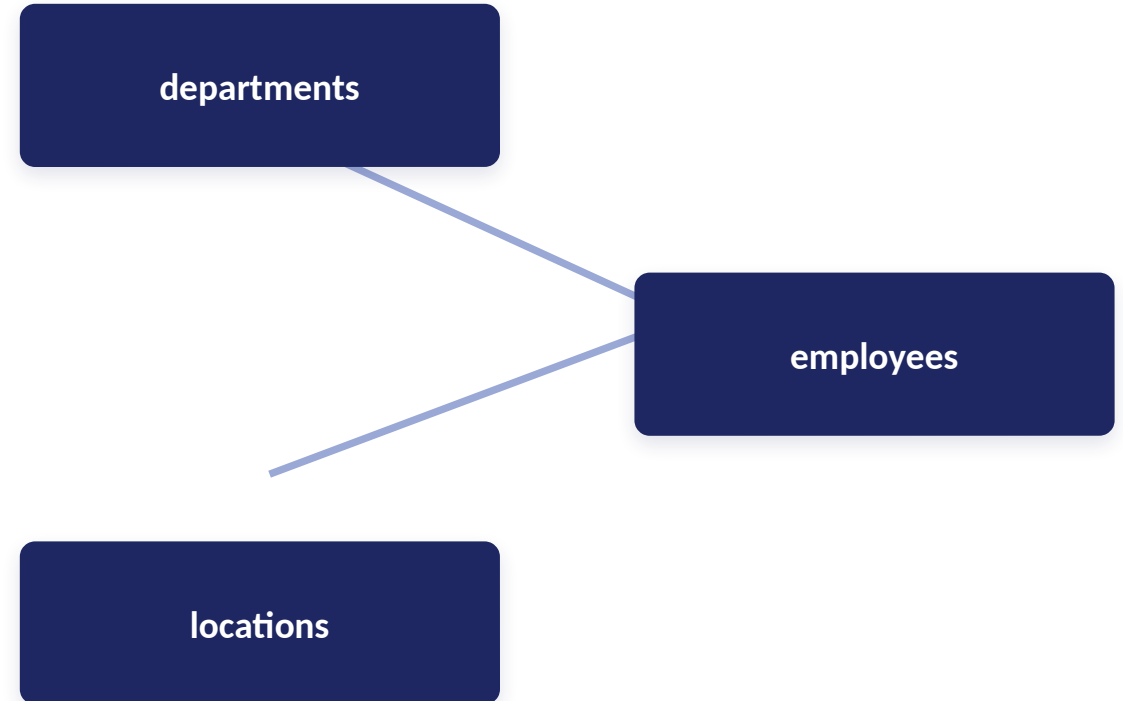
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What Is a Relational Database?

DEFINITION

A database that organizes data into separate tables (relations), each holding one type of entity, and links those tables together through shared key columns — instead of dumping everything into one giant sheet.



Tables linked by keys, not copy-pasted data



CONCEPT 05

What Is a Table?

DEFINITION

A structured grid of rows and columns that stores data about one type of thing — for example, the employees table stores one record per employee.

employee_id	first_name	department	salary
101	Ahmed	Engineering	92,000
102	Farah	Sales	68,500
103	Karim	Finance	77,250
104	Nadia	Marketing	61,000

This is Northstar's employees table — one row per person, one column per attribute.



What Is a Row? What Is a Column?

ROW = ONE RECORD

employee_id	first_name	department	salary
101	Ahmed	Engineering	92,000
102	Farah	Sales	68,500
103	Karim	Finance	77,250

A row (record) holds every value that belongs to one single employee — Farah's full record, end to end.

COLUMN = ONE ATTRIBUTE

employee_id	first_name	department	salary
101	Ahmed	Engineering	92,000
102	Farah	Sales	68,500
103	Karim	Finance	77,250

A column (field) holds one attribute — salary — for every employee, so values in it are always the same type.



CONCEPT 08

What Is a Primary Key?

DEFINITION

A column (or set of columns) that uniquely identifies every row in a table. It can never be NULL and never repeats — every `employee_id` belongs to exactly one employee.

- Must be unique
- Must never be NULL
- Usually auto-generated (e.g. `AUTO_INCREMENT`)

`employee_id` is the primary key

employee_id	first_name	department
101	Ahmed	Engineering
102	Farah	Sales
103	Karim	Finance



PK



CONCEPT 09

What Is a Foreign Key?

DEFINITION

A column in one table that points to the primary key of another table. It's how two tables get linked — the value must already exist as a primary key somewhere else.





CONCEPT 10

What Is a Relationship?

DEFINITION

The connection between two tables, formed through a primary key / foreign key pair — it comes in three shapes: one-to-one, one-to-many, and many-to-many.

1 : 1

One-to-One

One employee ↔ one payroll profile.

EXAMPLE

employee_id	name
101	Ahmed

▼ 1 : 1

employee_id	monthly_pay
101	92,000

Employee 101 has exactly one payroll row — never zero, never two.

1 : N

One-to-Many

One department → many employees.

EXAMPLE

department_id	name
1	Engineering

▼ 1 : N

employee_id	department_id
201	1
202	1
203	1

Department 1 appears once, but its id repeats on 3 employees.

N : N

Many-to-Many

Employees ↔ projects, both repeat.

EXAMPLE

employee_id	project_id
301	501
301	502
302	501
302	502

E301 & E302 both link to P501 and P502 — both columns repeat, so neither side is unique.



What Is Normalization?

DEFINITION

The process of organizing tables to remove repeated data and reduce inconsistency — typically applied in stages called normal forms (1NF, 2NF, 3NF).

BEFORE — REPEATED DATA

employee	department	dept_budget
Ahmed	Engineering	2,400,000
Farah	Engineering	2,400,000
Karim	Engineering	2,400,000

dept_budget repeats on every row — update it once, forget a row, and the data disagrees with itself.

AFTER — NORMALIZED

employee	department_id	department_id	budget
Ahmed	1	1	2,400,000
Farah	1		
Karim	1		

The budget lives once, in departments — employees just reference it by id.

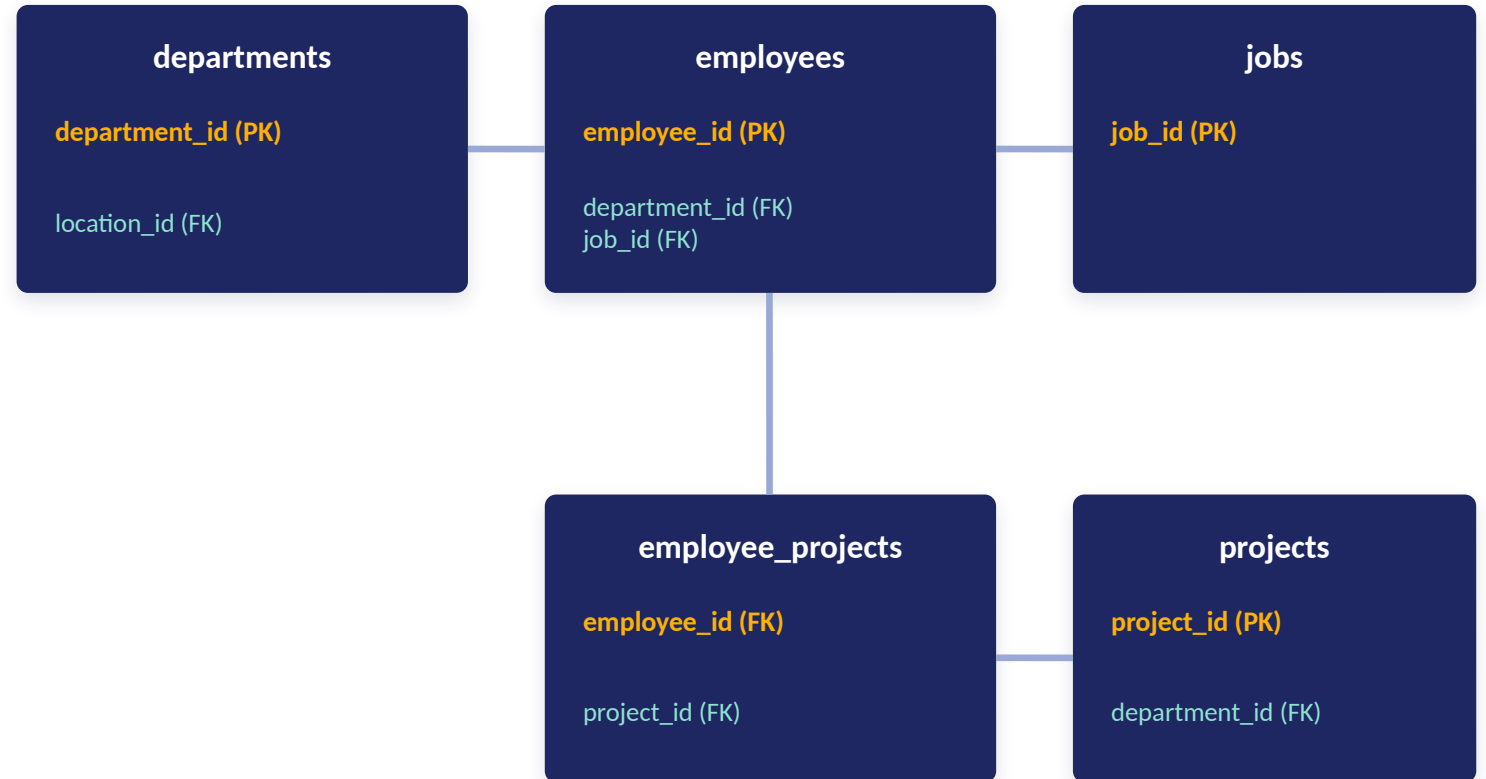


CONCEPT 12

What Is an ERD?

DEFINITION

An Entity Relationship Diagram is a visual blueprint of a database — boxes for tables (entities), their key columns, and lines showing how they relate. It's the map you draw before you build.



Every relationship you learned on the last two slides — PK, FK, and 1:N / N:N — is exactly what an ERD draws out visually.



CONCEPT 13

Why Do Businesses Use SQL?

1

Universal Standard

The same language works across MySQL, PostgreSQL, SQL Server, Oracle, and more.

2

Speed at Scale

Filters, joins, and aggregates millions of rows far faster than spreadsheets.

3

Reliable & Consistent

Enforces keys and types so many users can read/write the same data safely.

4

Powers Reporting

Feeds dashboards, KPIs, and BI tools directly from live business data.

5

One Skill, Every Role

Analysts, engineers, marketers, and finance teams all query the same way.

6

Direct Answers

Ask a precise business question, get a precise, reproducible answer.



Where Does SQL Fit in the Analytics Lifecycle?



SQL isn't one stop on the pipeline — it's the connective tissue.

It pulls raw rows out of storage, reshapes and cleans them, and rolls them up into the numbers a chart or a CEO actually reads — every single time data needs to move or answer a question.



YOU NOW KNOW

14 Concepts, Locked In

- ✓ Data
- ✓ DBMS
- ✓ Table
- ✓ Column
- ✓ Foreign key
- ✓ Normalization
- ✓ Why SQL
- ✓ Database
- ✓ Relational database
- ✓ Row
- ✓ Primary key
- ✓ Relationship
- ✓ ERD
- ✓ SQL in the analytics lifecycle

Next up: SELECT, WHERE, and ORDER BY — putting this vocabulary to work on Northstar's real employee data.